CHAPTER -1

INTRODUCTION

In recent years, the concept of smart agriculture has gained significant importance due to the increasing need for efficient resource management and reduced human effort. Traditional methods of plant watering require continuous manual monitoring, which can lead to overwatering or underwatering. To overcome these problems, automated systems using modern technology are being developed.

The **Internet of Things (IoT)** plays a crucial role in connecting physical devices to the internet, enabling remote monitoring and control. By using IoT, various environmental parameters such as soil moisture, temperature, and humidity can be continuously monitored in real time. This helps in maintaining optimal conditions for plant growth.

Moreover, in this project, the system is developed using the **ESP8266 microcontroller**, which has built-in Wi-Fi capability. It collects data from sensors like soil moisture sensor and DHT11 sensor and processes it accordingly. Based on the sensor values, the system can automatically control irrigation without human intervention.

The system also integrates with the **Blynk application**, which provides a user-friendly interface to monitor plant conditions remotely. Users can view real-time data and also manually control the water pump through the mobile application from anywhere.

This method not only saves time but also avoids manual errors in watering. It ensures healthy plant growth, minimizes the chances of diseases caused by improper watering, and is ideal for indoor plants, gardens, balconies, and even small farms. The system is user-friendly, low-cost, and energy-efficient, making it suitable for hobbyists, students, and professionals.

CHAPTER-2
LITERATURE REVIEW

In recent years, many researchers have focused on developing smart irrigation systems using IoT technology. These systems aim to reduce water wastage and improve plant growth by automating the irrigation process. Traditional irrigation methods were found to be inefficient due to lack of proper monitoring and control.

A Several studies have used soil moisture sensors to determine the water requirement of plants. Based on the sensor readings, automated irrigation systems were designed to supply water only when necessary. This approach has proven effective in conserving water and maintaining proper soil conditions.

Researchers have also integrated temperature and humidity sensors like DHT11 or DHT22 to monitor environmental conditions. These parameters play an important role in plant growth, and maintaining them within optimal limits helps in improving crop yield and health.

A With the advancement of wireless communication, microcontrollers such as Arduino and ESP8266 have been widely used in IoT-based projects. The ESP8266, in particular, is popular due to its built-in Wi-Fi capability, low cost, and ease of integration with sensors and cloud platforms.

In Many existing systems use mobile applications such as the **Blynk platform** to provide real-time monitoring and control. These applications allow users to access data remotely and control devices like water pumps, making the system more flexible and user-friendly.

Overall, previous research shows that IoT-based plant monitoring systems are effective in automating irrigation, reducing human effort, and improving water efficiency. However, there is still a need for simple, low-cost, and user-friendly solutions for small-scale users, which this project aims to address.

CHAPTER-3

PROPOSED SYSTEM

The proposed is an IoT-based smart plant monitoring and control system designed to automate irrigation and monitor environmental conditions. It uses an ESP8266 microcontroller to collect data from a soil moisture sensor and DHT11 sensor. When the soil moisture level falls below a set threshold, a water pump is automatically activated using a relay module, ensuring proper watering. The system is connected to the Blynk application via Wi-Fi, allowing users to monitor temperature, humidity, and soil moisture in real time. Users can also manually control the pump, making the system efficient and user-friendly.

Working Overview:

The system usesThe system uses ESP8266 to read data from soil moisture and DHT11 sensors and continuously monitors plant conditions. When the soil moisture level becomes low, the relay automatically turns ON the water pump to irrigate the plant. The collected data is sent to the Blynk application via Wi-Fi for real-time monitoring. Users can also manually control the pump through the mobile app, making the system efficient and flexible.

Key Functionalities:

1. Soil Monitoring: Continuously checks soil moisture using a sensor.
2. Automatic Control: Waters the plant when moisture drops below a set level.
3. Relay Operation: Uses a relay module to switch the water pump on/off.
4. Water Efficiency: Reduces water wastage by watering only when needed.
5. Low Maintenance: Requires minimal human intervention once set up**.**

Advantages of the Proposed System:

- Home gardens: Simplifies the care of indoor and outdoor plants.
- Agriculture: Used in large-scale forming to optimize water usage and crop yield.
- Greenhouses: Maintains optimal growing conditions for plants.
- Potted plants: Ideal for busy individuals who want to maintain plant health without daily attention.

<u>Future Scope:</u>

- Solar-powered system.
- Water conservation techniques.
- Pest and disease detection.
-

3.1 BLOCK DIAGRAM

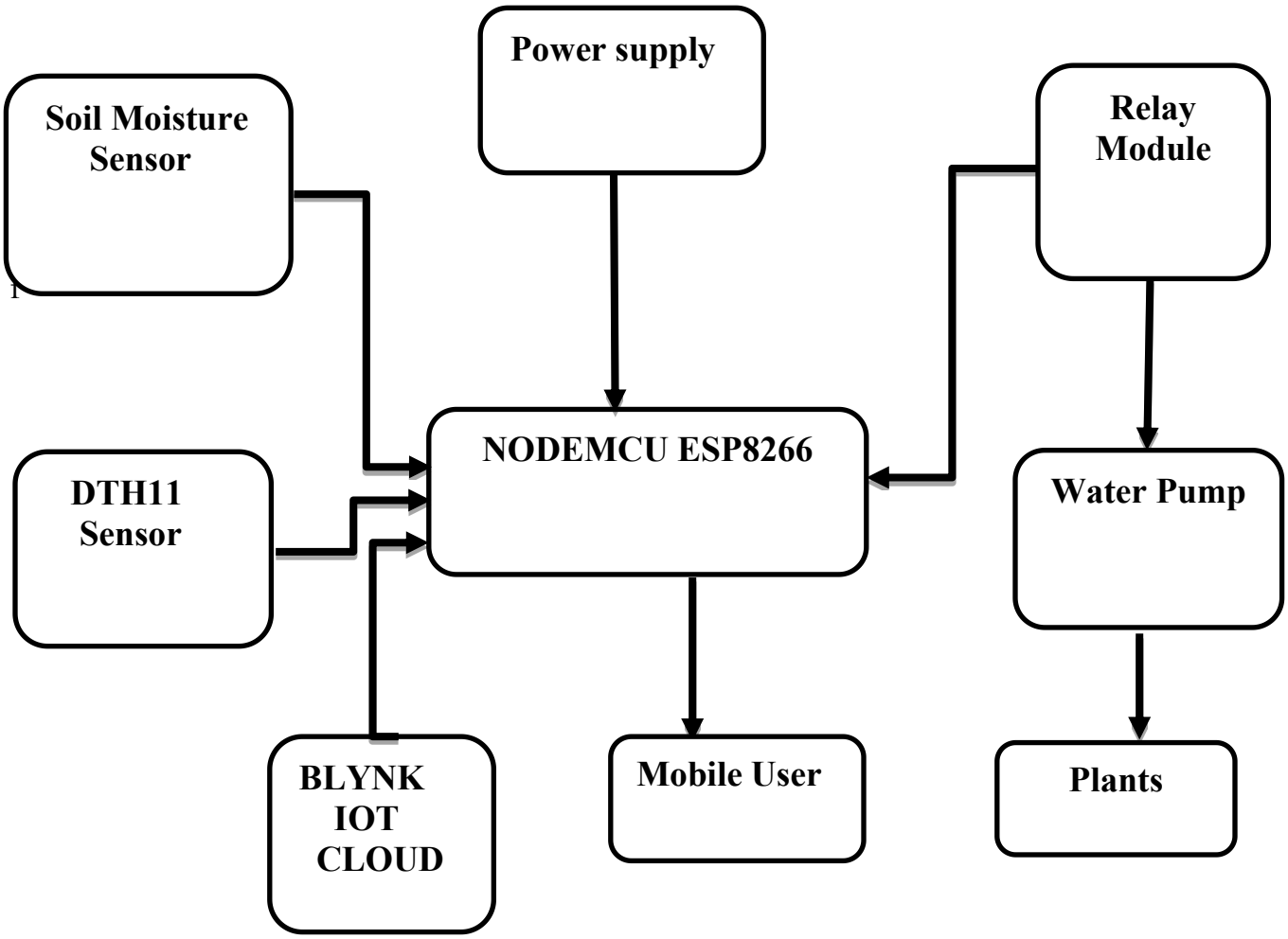


Figure 3.1: Block diagram

3.2 HARDWARE AND SOFTWARE DESCRIPTION

3.2.1 NODEMCU (ESP8266)

The NodeMCU is an open-source IoT development board based on the ESP8266 Wi-Fi module. It is widely used for building IoT applications because of its low cost, small size, and built-in Wi-Fi capability. The ESP8266 microcontroller allows devices to connect to the internet and communicate with other devices. NodeMCU comes with GPIO pins, which can be used to connect sensors, relays, and other electronic components. It supports serial communication protocols like UART, SPI, and I2C, making it highly flexible for various applications. The board operates at 3.3V and has an inbuilt voltage regulator, which makes it easy to use with external power sources. It also includes a USB-to-serial converter, allowing easy programming through a computer.

The NodeMCU can be programmed using the Arduino IDE, which makes it beginner-friendly and easy to develop projects. It supports Embedded C, Lua scripting, and other programming languages. The ESP8266 has a powerful Tensilica Xtensa 32-bit processor, which provides good processing speed for IoT applications. It also has flash memory for storing programs and data.

While NodeMCU is widely used in smart applications like home automation and plant monitoring. It can send data to Blynk for real-time monitoring and remote control.

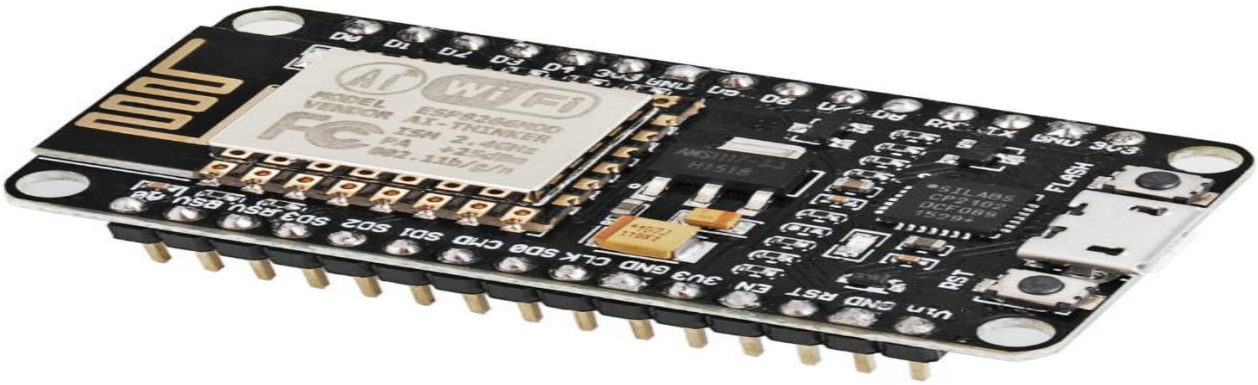


Figure 3.2.1: NODEMCU (ESP8266)

Technical Specifications:

Microcontroller: ESP8266
CPU: Tensilica Xtensa LX106
Architecture: 32-bit
Clock Speed: 80 MHz
Overclock Speed: 160 MHz
Operating Voltage: 3.3V
Input Voltage (Vin): 5V–12V
SPI: 1
Flash Memory: 4MB
SRAM: 64KB
GPIO Pins: 17
Digital I/O: Supported
Analog Input: 1 pin
ADC Resolution: 10-bit
Wi-Fi Standard: 802.11 b/g/n
Wi-Fi Frequency: 2.4 GHz
TCP/IP Stack: Supported
USB Interface: Micro USB
USB to Serial: CP2102 / CH340
Power Sources: DC Power Jack, USB Port and the VIN pin

Interrupt Support: Yes
Reset Button: Available
Flash Button: Available
Antenna: Onboard PCB antenna
Power Consumption: Low
Deep Sleep Mode: Supported
Wake-up Function: Supported
Operating Temperature: -40°C to 125°C
Wi-Fi Range: ~100m (open area)
Security: WPA/WPA2
Programming: Arduino IDE
Programming Language: Embedded C

3.2.2General pin functions:

1. **GPIO Pins (D0–D8):** Used for connecting sensors, relays, and other devices (input/output).
2. **D0 (GPIO16):** Used for deep sleep wake-up function.
3. **D1 (GPIO5):** Provide Used as SCL (clock) in I2C communication.
4. **D2 (GPIO4):** Supply Used as SDA (data) in I2C communication.
5. **D3 (GPIO0):** Ground Used for boot mode selection (LOW for programming mode).
6. **D4 (GPIO2):** Used Connected to the built-in LED.
7. **D5 (GPIO14):** Allows Used as SCLK in SPI communication.
8. **D6 (GPIO12):** Used as MISO in SPI communication.
9. **D7 (GPIO13):** Used as MOSI in SPI communication.
10. **D8 (GPIO15):** Used as CS (Chip Select) in SPI communication.
11. **RX (GPIO3):** Serial receive pin.
12. **TX (GPIO1):** Serial transmit pin.
13. **A0:** Analog input pin for reading sensor values.
14. **3V3:** 3.3V power output.
15. **Vin:** External power input (5V).
16. **GND:** Ground connection.

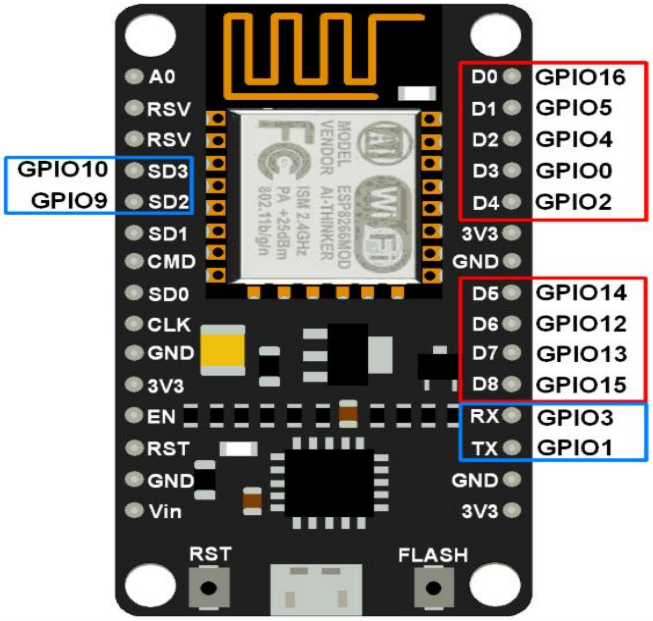


FIGURE 3.2.2: NODEMCU(ESP8266) pin description

Special pin functions

The NodeMCU (ESP8266) includes several special function pins that perform specific tasks. Pins like D1 and D2 are used for I2C communication, while D5 to D8 are used for SPI communication. The TX and RX pins handle serial communication for data transfer. D0 is used for deep sleep wake-up functionality, helping in power saving. D3 and D4 are important for boot mode selection and include the built-in LED. The A0 pin is used for analog input to read sensor values. These special functions make the NodeMCU flexible and suitable for various IoT applications and smart systems.

NodeMCU (ESP8266) has multiple pins used for different functions. The digital pins (D0–D8) are used as input/output to connect sensors, relays, and other devices. The A0 pin is used for analog input to read sensor values, while TX and RX pins are used for serial communication. The 3V3 pin provides 3.3V output, Vin is used for external power input, and GND pins are used for grounding. In addition, some pins have specialized functions such as I2C, SPI, and deep sleep control, making the board more flexible for IoT applications.

3.2.3 Arduino IDE Software

The Arduino IDE is used to write and upload the code for the IoT based smart plant monitoring and control system. It reads soil moisture levels using a sensor and controls the water pump through a relay. The IDE allows easy programming and debugging with real-time serial monitor output. Once uploaded, the Arduino automatically manages watering based on soil moisture.

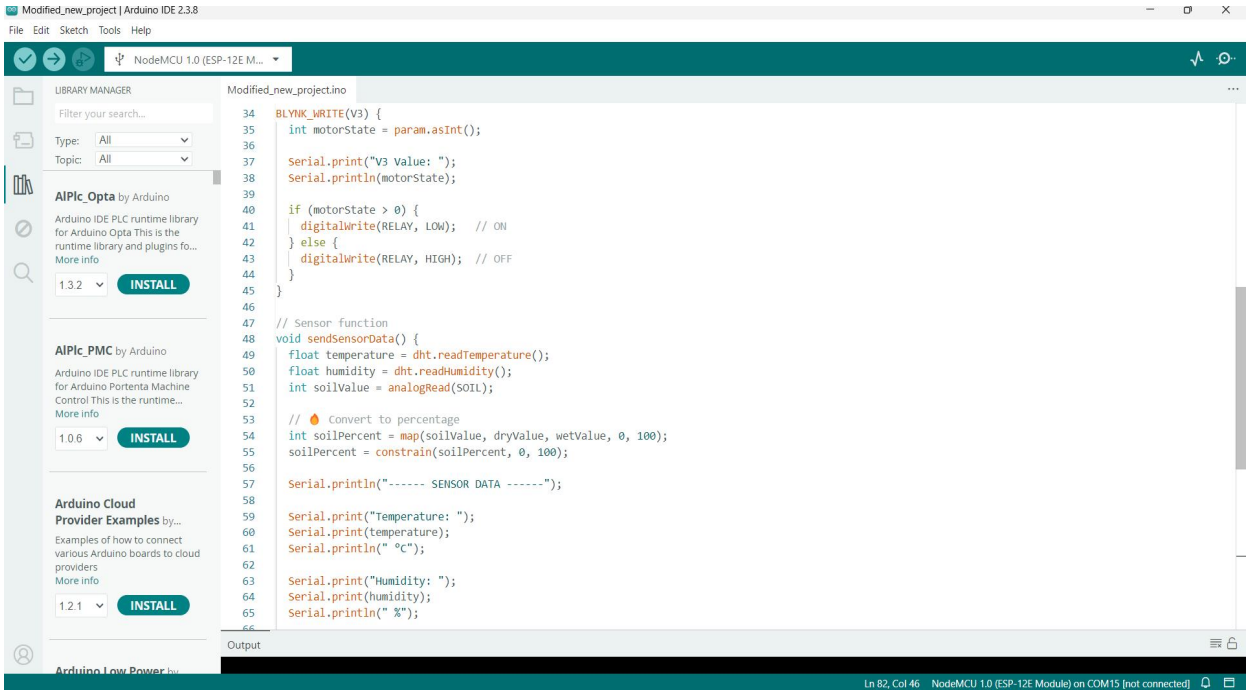


Figure 3.2.3: Arduino IDE

Legacy IDE

Arduino integrated development environment (IDE) is a cross-platform application (for Microsoft Windows, macOS, and Linux) that is written in the Java programming language. It originated from the IDE for the languages Processing and Wiring. It includes a code editor with features such as text cutting and pasting, searching and replacing text, automatic indenting, brace matching, and syntax highlighting,

And provides simple one-click mechanisms to compile and upload programs to an Arduino board. It also contains a message area, a text console, a toolbar with buttons for common functions and a hierarchy of operation menus. Thesource code for the IDE is released under the GNU General Public License, version 2.

The Arduino IDE supports the languages C and C++ using special rules of code structuring. The Arduino IDE supplies a software library from the Wiring project, which provides many common input and output procedures. User-written code only requires two basicfunctions, for starting the sketch and the main program loop, that are compiled and linked witha program stub main into an executable cyclic executive program with the GNU toolchain.

The Arduino IDE employs the program argued to convert the executable code into a text file in hexadecimal encoding that is loaded into the Arduino board by a loader program inthe board's firmware.

From version 1.8.12, Arduino IDE windows compiler supports only Windows 7 or newer OS. On Windows Vista or older one gets "Unrecognized Win32 application" error whentrying to verify/upload program. To run IDE on older machines, users can either use version 1.8.11, or copy "arduino-builder" executable from version 11 to their current install folder as it's independent from IDE.

Sketch:

A sketch is a program written with the Arduino IDE. Sketches are saved on the development computer as text files with the file extension,ino. Arduino Software (IDE) pre-1.0 saved sketches with the extension,pde.

A minimal Arduino C/C++ program consists of only two functions:

- setup: This function is called once when a sketch starts after power-up or reset. It is used to initialize variables, input and output pin modes, and other libraries needed in the sketch. It is analogous to the function main.
- loop :: After setup function exits (ends), the loop function is executed repeatedly in the main program. It controls the board until the board is powered off or is reset. It is analogous to the function while

3.2.4 Blink example:

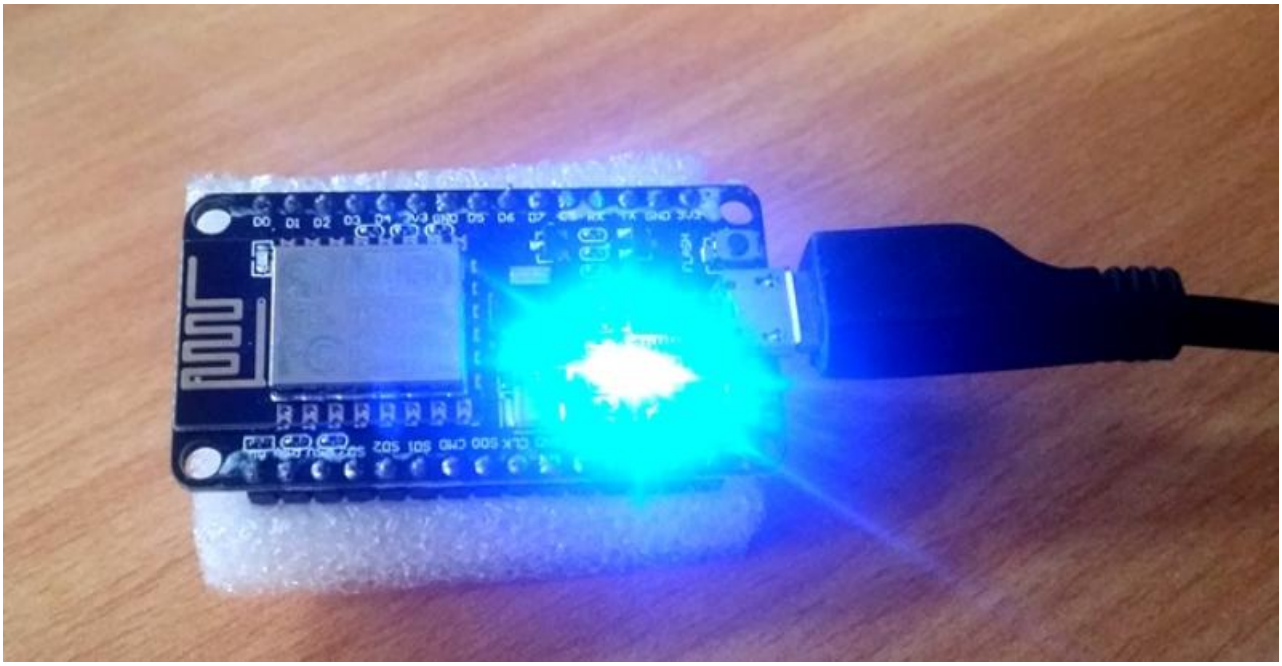


Figure 3.2.4: NODEMCU(ESP8266) LED blink

The NodeMCU ESP8266 board has a built-in blue LED connected to GPIO2 (D4 pin), which is commonly used for indication and testing purposes. This LED helps users verify whether the board is functioning correctly. When the pin is set to LOW, the LED turns ON, and when set to HIGH, it turns OFF due to inverted logic. The blue LED also blinks during booting and program uploading, indicating activity. It is very useful for beginners to test simple programs like LED blinking without requiring external components, making development easier and faster.

Libraries:

The open-source nature of the Arduino project has facilitated the publication of many free software libraries that other developers use to augment their projects.

Operating systems/threading:

There NodeMCU (ESP8266) does not run a full operating system but uses lightweight firmware. It follows a single-threaded model where tasks run sequentially in the loop function. However, it supports event-driven programming and handles Wi-Fi tasks in the background, giving a multitasking-like behavior without true multithreading support.

Applications:

- Arduboy, a handheld game console based on Arduino.
- Arduinome, a MIDI controller device that mimics the Monome.
- Ardupilot, drone software and hardware.
- ArduSat, a cubesat based on Arduino.
- C-STEM Studio, a platform for hands-on integrated learning of computing, science, technology, engineering, and mathematics (C-STEM) with robotics.
- Data loggers for scientific research .
- OBDuino, a trip computer that uses the on-board diagnostics interface found in most modern cars.
- OpenEVSE an open-source electric vehicle charger.
- XOD, a visual programming language for Arduino.

Simulation:

- Simulation helps to test the system design using software like Proteus before hardware implementation.
- It reduces errors, saves time, and improves accuracy by identifying issues early.

Recognitions:

The Arduino project received an honorary mention in the Digital Communities category at the 2006 Prix Ars Electronica. The Arduino Engineering Kit won the Bett Award for "Higher Education or Further Education Digital Services" in 2020.

3.2.5 Soil moisture Sensor:

A soil moisture sensor is a device that measures the water content in soil. It typically works by detecting changes in electrical resistance or the dielectric constant of the soil, which varies with moisture levels. The sensor consists of probes inserted into the soil, providing data on soil hydration. This information can be used to automate irrigation systems, ensuring plants receive adequate water while conserving resources. Soil moisture sensors are widely used in agriculture, gardening, and greenhouses to monitor soil health, prevent overwatering or under watering, and optimize irrigation for efficient water usage and healthier plant growth.

Figure 3.2.5: Soil moisture sensor

Working Principle:

1. Resistance Measurement: The sensor measures electrical resistance between two probes in the soil. More moisture lowers resistance, while dry soil increases it.

2. Voltage Output: The sensor outputs an analog voltage based on soil moisture level. Higher moisture gives lower voltage, and dry soil gives higher voltage.

3. Capacitive Sensing : Capacitive sensors measure changes in the soil's dielectric constant. Water increases the dielectric value, which is detected by the sensor.

4. Sensor Probes: Two metal probes are inserted into the soil to sense conductivity. Wet soil conducts electricity better than dry soil.

5. Signal Conversion: The analog signal is converted to digital using a microcontroller. This helps in precise monitoring and automation control.

6. Automation Trigger: Based on the moisture level, the system triggers a pump or alert This helps automate irrigation only when needed.

Specifications:

Knowing the specifications of an soil moisture sensor helps in understanding the reliable approximations of soil measurement.

1. Operating Voltage: 3.3V – 5V DC

2. Output Type: Analog and Digital

3. Current Consumption: < 20 mA

4. Sensor Type: Resistive or Capacitive

5. Probe Material: Usually corrosion-resistant metal

6. Measurement Range: 0% (dry) to 100% (wet) soil moisture

7. Analog Output Voltage: 0 – 4.2V (depends on moisture)

8. Digital Output: High/Low based on set threshold

9. PCB Size: Around 3.5cm x 1.5cm (varies by model)

10. Applications: Gardening, agriculture, greenhouse automation

Pin discription:

1. VCC: Connects to the 3.3V or 5V power supply. Powers the sensor module.

2. GND: Ground pin; connects to the GND of NODEMCUESP8266 or power source Completes the circuit.

3. A0 (Analog Output): Provides analog voltage based on soil moisture level**.**

4. D0 (Digital Output): Outputs HIGH or LOW based on a set moisture threshold. Can be used to trigger value devices.

3.2.6 Relay Module:

A relay module is an essential electronic component used to control high-voltage devices like lights, fans, or water pumps with low-voltage signals from microcontrollers such as Arduino. It works on the principle of electromagnetic induction. When a small current is applied to the relay's input pin, it energizes an internal coil that creates a magnetic field. This field pulls a switch inside the relay, either opening or closing the circuit connected to a high-voltage device. This allows microcontrollers to safely control devices that require more voltage or current than the controller can handle directly.

In an automatic plant watering system, the relay module plays a key role in switching the water pump ON and OFF based on soil moisture readings. If the soil is dry, the Arduino sends a signal to the relay, which then activates the pump to water the plant. When the soil becomes moist again, the relay turns the pump off. Relay modules also offer electrical isolation, which protects sensitive components from high voltage damage. With features like LED indicators, screw terminals, and multiple channels, relay modules are simple to use and reliable, making them perfect for automation and home gardening projects.

Figure 3.2.6: Relay module

Working principle:

A relay module works based on the principle of electromagnetism. When a low voltage signal (like 5V from an Arduino) is applied to the relay's input pin, it activates an electromagnetic coil inside the relay. This coil generates a magnetic field which pulls a metal switch (called an armature) to change its position.

In the normally open (NO) configuration, when the coil is energized, the switch closes, completing the circuit and allowing current to flow to the high-voltage device like a water pump or light. When the signal is turned OFF, the magnetic field disappears, and the switch returns to its original position, breaking the circuit and turning off the device.This allows the relay to safely control high voltage or current devices using a small control signal. It acts as an interface between the microcontroller and electrical appliances, providing electrical isolation and protection.

Key Components and Pin Configuration:

A Relay module consists of the following components:

1. **VCC:** Power pin, connected to 5V from Arduino.
2. **GND:** Ground pin, connected to Arduino's GND.
3. **IN (Input):** Signal pin, receives HIGH/LOW signal from Arduino to control the relay.
4. **COM (Common):** Common terminal for the relay switch, usually connected to the load.
5. **NO (Normally Open):** When relay is OFF, this pin is open. Closes when relay is activated.
6. **NC (Normally Closed):** Connected when relay is OFF; opens when relay is ON.

Features and Specifications:

1. Operating Voltage: 5V DC, powered directly.

2. High Voltage Support: Can switch high-voltage AC (up to 250V) or DC (up to 30V), making it suitable for controlling water pumps or solenoid valves.

3. Trigger Voltage: The relay can be triggered with a control voltage of 3.3V to 5V, compatible with Arduino's output.

4. Max Switching Current: Handles up to 10A, suitable for controlling heavy-duty devices.

5. LED Indicator: Shows the status of the relay (ON/OFF), providing visual feedback for debugging and monitoring.

6. Electrical Isolation: The relay provides electrical isolation between the low-voltage control circuit and the high-voltage device, ensuring safety**.**

7. Compact Design: Small in size, it's easy to integrate into the plant watering system and other automation projects.8. Connection Terminals: Equipped with screw terminals for easy and secure connections to the pump and power sources**.**

Conclusion:

The relay module is a crucial component in automation systems, such as the Automatic Plant Watering System. It allows microcontrollers like NODEMCUESP8266 to safely control high-voltage devices using low-voltage signals. With features like electrical isolation, high-voltage compatibility, and easy interfacing, the relay ensures safe and reliable switching of devices like water pumps. The module's compact design and LED indicator make it ideal for DIY and embedded projects. In conclusion, the relay module is essential for effective and secure control of high-power devices, contributing to the efficiency and safety of automated systems.

3.2.7 Motor /pump

In the Automatic Plant Watering System, the water pump plays a crucial role in ensuring that plants receive the required amount of water. The pump draws water from a source, such as a tank or reservoir, and transports it to the plant's roots. This process is controlled by a microcontroller (like NODEMCUESP8266), which monitors the soil moisture level through a soil moisture sensor. When the sensor detects dry soil, it sends a signal to the relay module, which then activates the pump to water the plants. Once the soil reaches the desired moisture

level, the system automatically deactivate.

Figure.3.2.7: Motor/Pump

There are different types of pumps used in such systems, with the submersible pump being one of the most common. This type of pump is submerged in water, and it's ideal for larger setups or when the water source is deep. Another option is the peristaltic pump, often used for more precise water flow control in smaller systems. It ensures that the water is delivered in a steady, controlled manner. When choosing a pump for the watering system, it's essential to ensure that the voltage and current ratings of the pump are compatible with the relay module to ensure safe and efficient operation**.**

3.2.8 HW- battery

The battery is an essential hardware component in any portable or semi-portable automatic plant watering system. It serves as the primary or backup power source, especially in locations where constant access to electricity is not available. A reliable battery ensures that the system continues to function effectively, providing water to plants as needed, even during power cuts or in outdoor environments like gardens or greenhouses where wiring for power might be difficult.

In this system, the battery powers critical components such as the NODEMCUESP8266 microcontroller, soil moisture sensor, relay module, and the water pump. The battery must provide a stable voltage and sufficient current to operate these components simultaneously. Typically, lithium-ion batteries, lead-acid batteries, or rechargeable LiPo (Lithium Polymer) batteries are used, depending on the scale and design of the system. For small to medium systems, a 12V lithium-ion battery is most common due to its balance of capacity, weight, and availability.

One of the major advantages of using a battery-powered system is portability. The system can be

installed in remote areas without the need for a direct AC power source. This makes it ideal for agricultural fields, rooftop gardens, or even small home setups. Another benefit is energy independence, especially if the battery is integrated with a solar charging system, allowing the battery to recharge using solar panels, thus making the watering system completely self-sustainable and eco-friendly.

Figure 3.2.8: HW- battery

3.2.9 Plant:

In the IoT based smart plant monitoring and control system, the plant is the main subject being monitored and cared for. The system is designed to ensure the plant receives the right amount of water by automatically checking the moisture level in the soil. When the soil gets dry, the system activates a pump to deliver water to the plant. This helps prevent overwatering or underwatering, ensuring healthy plant growth. It is especially useful for indoor plants, gardens, pots, or agriculture, where regular manual watering is difficult.

Figure 3.2.9: Plant

CHAPTER – 4

IMPLEMENTATION AND RESULT

4.1 Working Principle:

The IoT based smart plant monitoring and control system works on the principle of monitoring soil moisture and automating the watering process. A soil moisture sensor constantly checks the water level in the soil. When the moisture level drops below a certain threshold, the NODEMCU(ESP8266) receives this data and triggers a relay module, which in turn powers the water pump. The pump draws water from a reservoir and delivers it to the plant. Once the soil reaches sufficient moisture, the pump automatically switches off.

For implementation, all components— NODEMCU(ESP8266), relay, moisture sensor, and pump—were connected on a simple breadboard circuit. The system was programmed using Arduino IDE, and the moisture levels were calibrated for different soil types.

Results:

The IoT based smart plant monitoring and controlling system successfully maintained the required moisture level in the soil without manual watering. It accurately sensed the dryness of the soil and activated the pump only when needed. This helped in saving water, reducing human effort, and keeping the plants healthy. The system worked efficiently for different types of plants and soils, proving its reliability and usefulness for home gardens, pots, and small-scale agriculture. Overall, the project achieved its goal of creating a smart, automated watering solution.

4.1 Schematic Diagram:

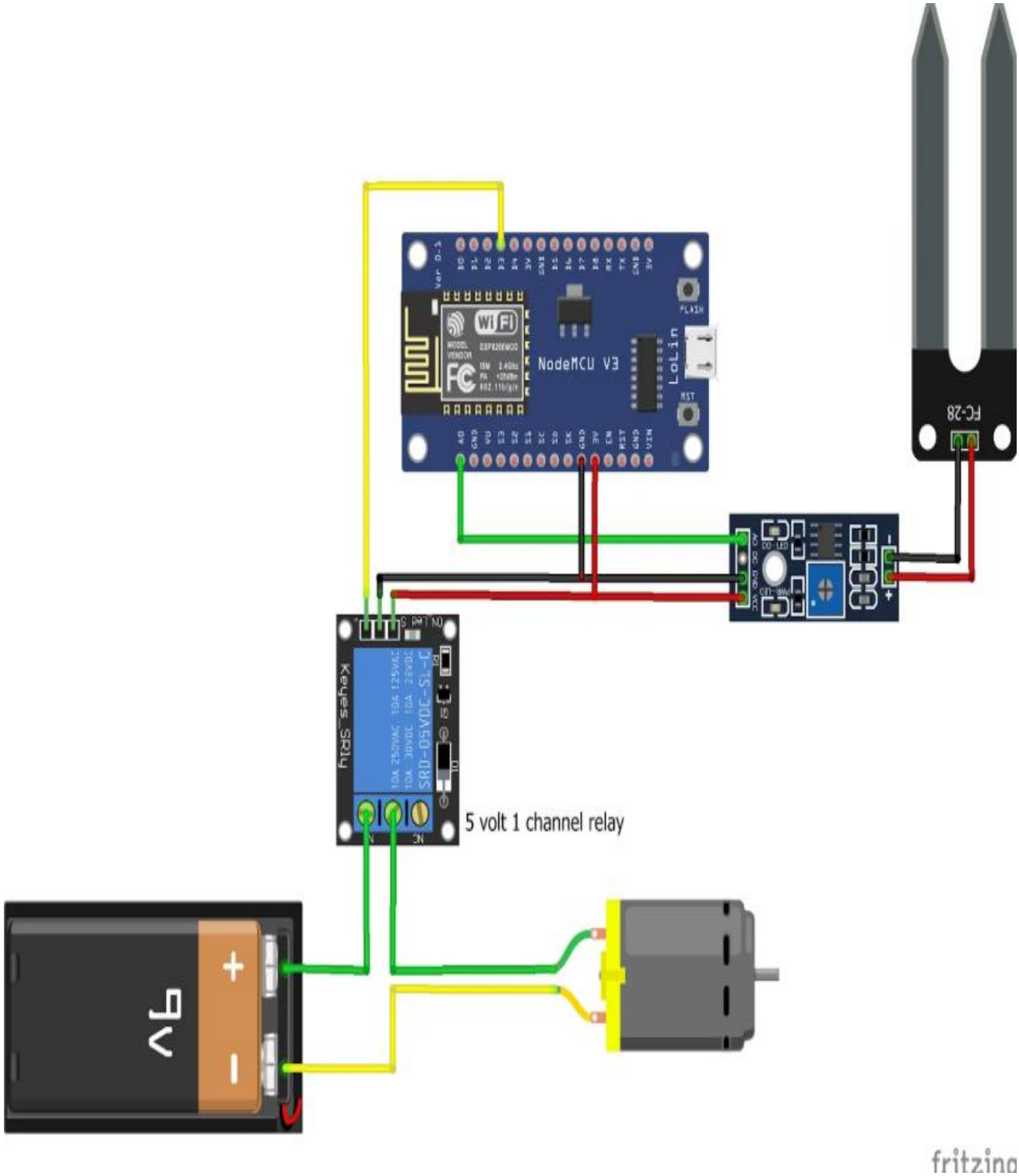


Figure 4.2: Schematic diagram

4.3 Flow Chart:

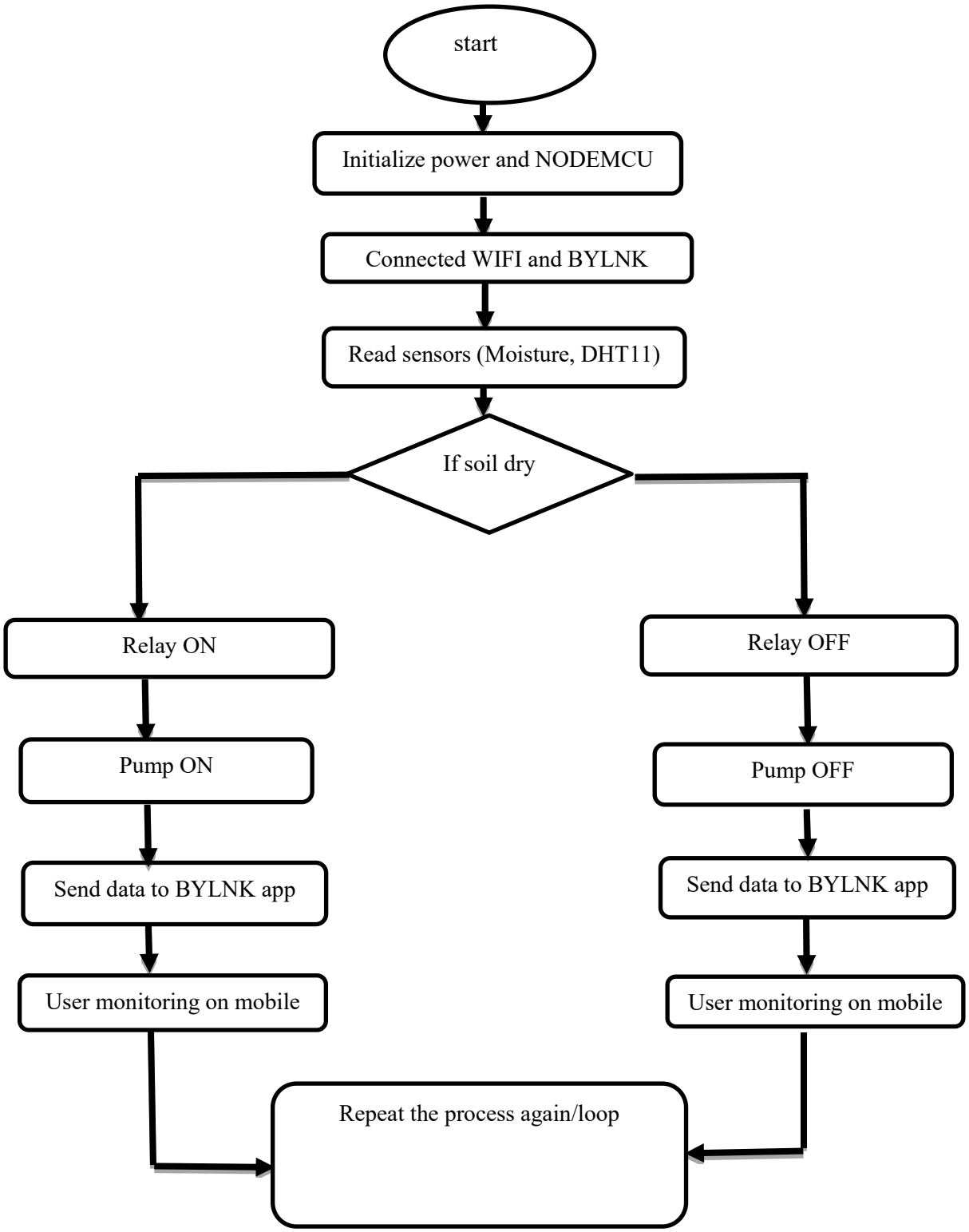


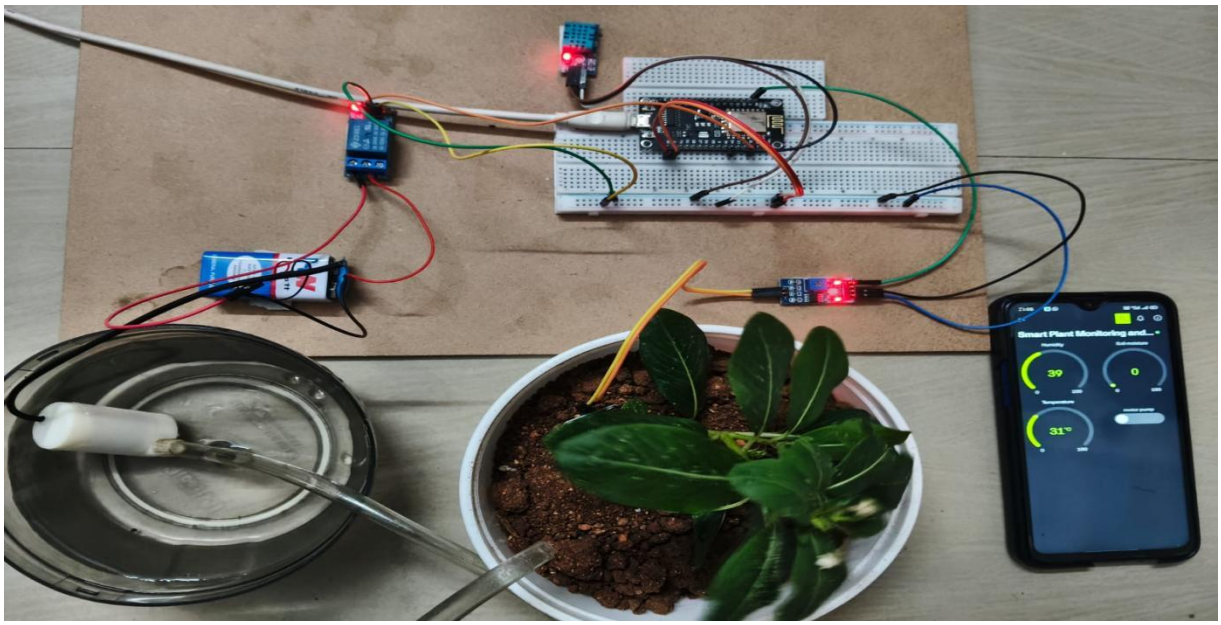
Figure 4.3: Flow chart

4.4 PROGRAM

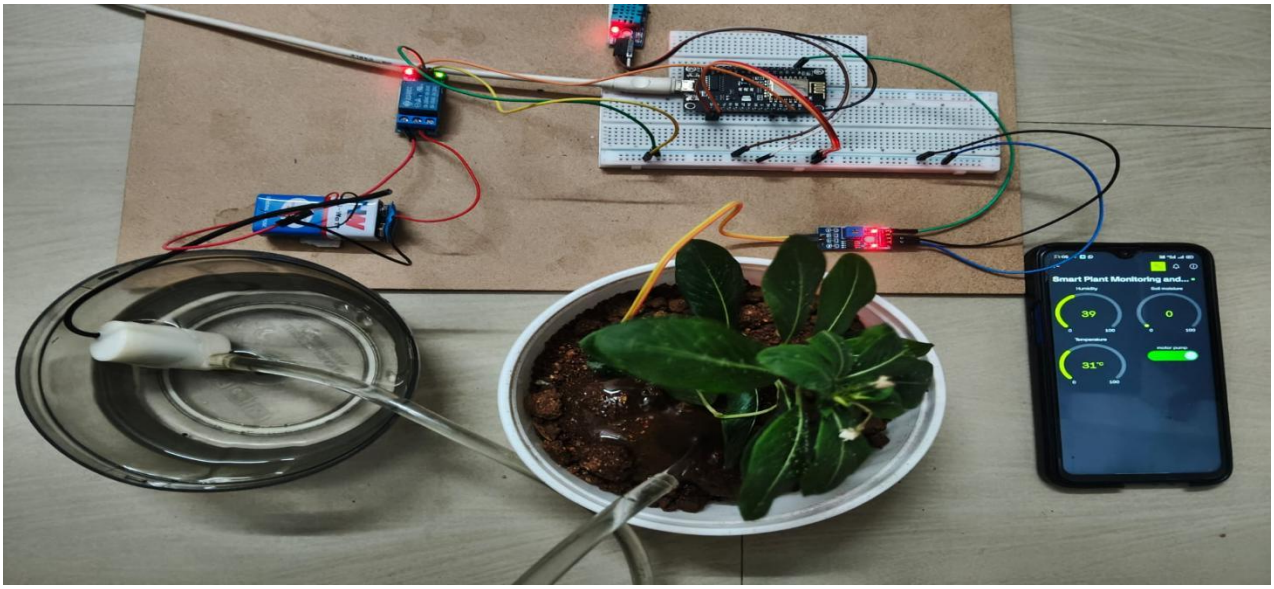
```
#define BLYNK_TEMPLATE_ID "TMPL3mmDXt35r"
#define BLYNK_TEMPLATE_NAME "Smart Plant Monitoring and control System"
#define BLYNK_AUTH_TOKEN "IPPxg9zbWuZUIqIn7iEvLyloOsufQzZn"

#define BLYNK_PRINT Serial

#include <ESP8266WiFi.h>
#include <BlynkSimpleEsp8266.h>
#include <DHT.h>

// WiFi
char ssid[] = "Bhanu";
char pass[] = "12345678";

// Pins
#define DHTPIN D4
#define DHTTYPE DHT11
#define RELAY D1
#define SOIL A0

DHT dht(DHTPIN, DHTTYPE);
BlynkTimer timer;

// 🔧 CALIBRATION VALUES (CHANGE IF NEEDED)
int dryValue = 700;   // Dry soil reading
int wetValue = 300;   // Wet soil reading

// Sync button state
BLYNK_CONNECTED() {
  Blynk.syncVirtual(V3);
}

// Motor control
BLYNK_WRITE(V3) {
  int motorState = param.asInt();

  Serial.print("V3 Value: ");
  Serial.println(motorState);

  if (motorState > 0) {
    digitalWrite(RELAY, LOW);   // ON
  } else {
    digitalWrite(RELAY, HIGH);  // OFF
  }
}

// Sensor function
void sendSensorData() {
  float temperature = dht.readTemperature();
  float humidity = dht.readHumidity();
  int soilValue = analogRead(SOIL);
```

```
  // 💧 Convert to percentage
  int soilPercent = map(soilValue, dryValue, wetValue, 0, 100);
  soilPercent = constrain(soilPercent, 0, 100);

  Serial.println("------ SENSOR DATA ------");

  Serial.print("Temperature: ");
  Serial.print(temperature);
  Serial.println(" °C");

  Serial.print("Humidity: ");
  Serial.print(humidity);
  Serial.println(" %");

  Serial.print("Soil Moisture (%): ");
  Serial.println(soilPercent);

  Serial.println("-------------------------");

  // Send to Blynk
  Blynk.virtualWrite(V0, temperature);
  Blynk.virtualWrite(V1, humidity);
  Blynk.virtualWrite(V2, soilPercent);
}

void setup() {
  Serial.begin(115200);

  pinMode(RELAY, OUTPUT);
  digitalWrite(RELAY, HIGH); // OFF initially

  dht.begin();

  Blynk.begin(BLYNK_AUTH_TOKEN, ssid, pass);

  timer.setInterval(3000L, sendSensorData);
}

void loop() {
  Blynk.run();
  timer.run();
}
```

4.5 RESULT



Before



After

Figure 4.5: Result

4.6 Advantages:

- Used in home gardening to automate plant watering.
- Helps farmers irrigate small agriculture fields efficiently.
- Maintains ideal moisture in greenhouse plants.
- Water rooftop or terrace garden plant automatically.
- Keeps office indoor plants automatically.
- Supports greenery maintenance in schools and colleges.
- Provide consistent watering in plant nurseries.
- Can automates watering in plant nurseries.
- Use in smart irrigation with IoT integration.

4.7 Applications:

- Used in home gardens to automate daily water.
- Ideal for terrace and balcony gardens in cities.
- Maintains moisture levels in greenhouse farming.
- Helps in small agricultural fields for smart irrigation.
- Keeps office plants healthy without manual efforts.
- Support school and college garden maintenances.
- Ensures regular watering in plant nurseries.
- Useful in public parks for automatic watering.
- Integrated into IoT systems for smart control.
- Perfect for watering plants in vacation homes.

CONCLUSION

The IoT based smart plant monitoring and control system is a groundbreaking solution that eliminates the need for constant manual watering of plants. By integrating simple yet effective components such as the NODEMCUESP8266 microcontroller, soil moisture sensor, relay module, and a water pump, the system can automatically detect when a plant requires water based on real-time soil moisture levels. This automation reduces the need for constant monitoring and ensures plants are watered precisely when needed, leading to healthier growth and efficient use of resources.

One of the most significant advantages of the system is its ability to conserve water. Traditional watering methods often lead to overwatering or under watering, which can harm plants and waste resources. The automatic watering system, however, delivers water only when the soil moisture drops below a certain threshold. This ensures that plants receive the exact amount of water they require, improving plant health and reducing water wastage, making it an environmentally sustainable solution.

FUTURESCOPE

1. Solar Power Support – Makes the system eco-friendly and power independent.

2. Scalability – Can be expanded from small gardens to large agricultural fields.

3. Automated Nutrient Supply – Combines watering with fertilizer distribution for better plant growth.

REFERENCES

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- "Automatic Irrigation System Using Arduino" – International Journal of Engineering Research
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